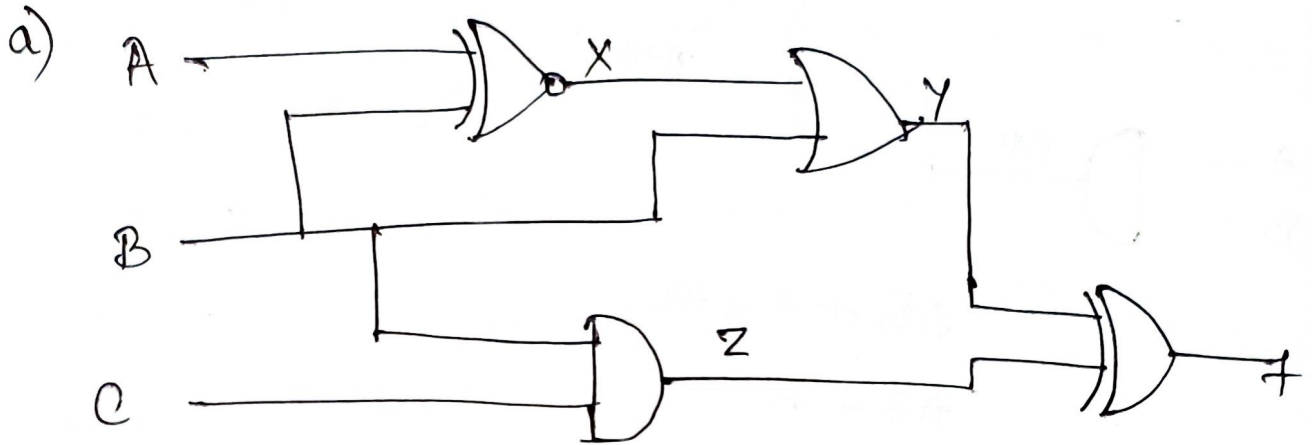


## Day-2: Assignment

- 1) Derive the output expression 'f' for the below circuits and draw the simplified circuits.



$$A \oplus B' = AB + A'B' = X$$

$$Y = X + B$$

$$= AB + A'B' + B$$

$$AB + B = B$$

$$Z = BC$$

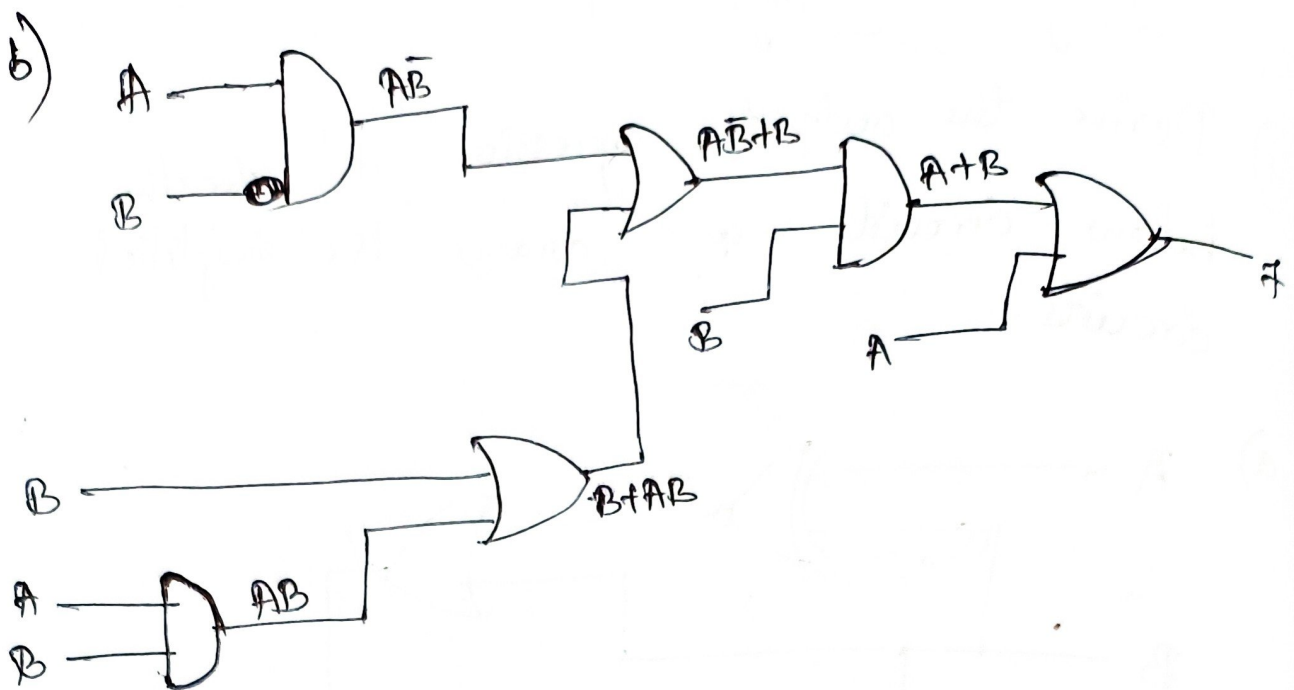
$$Y = B + A'B'$$

$$= B + A'B' = (B + A')(B + B') = B + A'$$

$$Y = A' + B$$

$$f = (A' + B) \oplus BC$$

$$= \underline{\underline{A'B' + A'C' + BC'}}$$



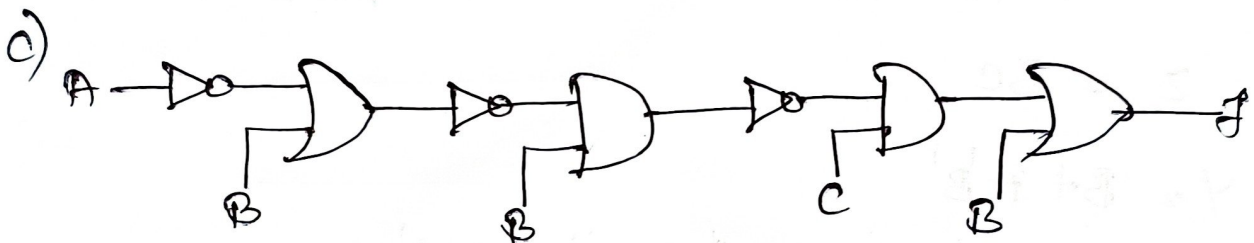
$$A\bar{B} + B + AB$$

$$A\bar{B} + B$$

$$B(A\bar{B} + B) = A + B$$

$$A + A + B$$

$$\underline{F = A + B}$$



$$AB' \Rightarrow (AB')B = 0 = 1$$

$$\boxed{F = C + B}$$

2) Minimise the following Boolean Expressions using Boolean postulates.

a)  $\bar{A}\bar{C}\bar{D}E + \bar{A}\bar{B}\bar{D} + ABCE + ABD$

$$= \bar{A}\bar{C}\bar{D}E + \bar{A}\bar{B}\bar{D} + AB(CE + D)$$

$$= \bar{A}\bar{D}(CE + \bar{B}) + AB(CE + D) \quad \left. \vphantom{\begin{aligned} &\bar{A}\bar{C}\bar{D}E + \bar{A}\bar{B}\bar{D} + AB(CE + D) \\ &= \bar{A}\bar{D}(CE + \bar{B}) + AB(CE + D) \end{aligned}} \right\} \times$$

$$= CE(\bar{A}\bar{D} + AB) + \bar{A}\bar{B}\bar{D} + ABD$$

$$= CE(AB + B\bar{D} + \bar{A}\bar{D}) + \bar{A}\bar{B}\bar{D} + ABD$$

$$= ABCE + B\bar{D}CE + \bar{A}\bar{D}CE + \bar{A}\bar{B}\bar{D} + ABD$$

$$= AB(CE + D) + B\bar{D}CE + \bar{A}\bar{D}CE + \bar{A}\bar{B}\bar{D}$$

$$= \underline{ABD + BC\bar{E}\bar{D} + \bar{A}\bar{B}\bar{D}}$$

$$b) \bar{A}\bar{B}\bar{C} + ABD + \bar{A}C + \bar{A}C\bar{D} + \bar{A}C\bar{D} + \bar{A}\bar{B}\bar{C}$$

$$\bar{A}C + \bar{A}C\bar{D} = \bar{A}C$$

$$= \bar{A}\bar{B}\bar{C} + ABD + \bar{A}C + \bar{A}C\bar{D} + \bar{A}\bar{B}\bar{C}$$

$$= \bar{B}\bar{C}(\bar{A} + A) + ABD + \bar{A}C + \bar{A}C\bar{D}$$

$$= \bar{B}\bar{C} + ABD + \bar{A}C + \bar{A}C\bar{D}$$

$$= \bar{B}\bar{C} + \bar{A}C + AD(B + \bar{C})$$

$$= \underline{\bar{B}\bar{C} + \bar{A}C + AD(\bar{C} + B)}$$

$$c) \bar{A}\bar{C} + BC + \bar{A}\bar{B} + \bar{A}BC + \bar{A}C\bar{D} + \bar{B}\bar{C}\bar{D}$$

$$\bar{A}\bar{C} + BC(\bar{A} + 1) + \bar{A}C\bar{D} + \bar{B}\bar{C}\bar{D}$$

$$\bar{A}\bar{C} + BC + \bar{A}C\bar{D} + \bar{B}\bar{C}\bar{D}$$

$$(AC + \bar{A}\bar{C})\bar{D} + (BC + \bar{B}\bar{C})\bar{D}$$

$$\underline{\bar{A}\bar{C} + BC + \bar{A}\bar{B}}$$

$$d) \bar{x}_1\bar{x}_2\bar{x}_4 + x_3x_4 + \bar{x}_1\bar{x}_2x_4 + x_1x_2\bar{x}_3x_4 + x_2$$

$$\bar{x}_1\bar{x}_2\bar{x}_4 + x_3x_4 + x_1x_2\bar{x}_3x_4 + \bar{x}_1x_4(x_2 + \bar{x}_2)$$

$$\bar{x}_4 + \bar{x}_1x_4 + x_3x_4 + x_1x_2\bar{x}_3x_4$$

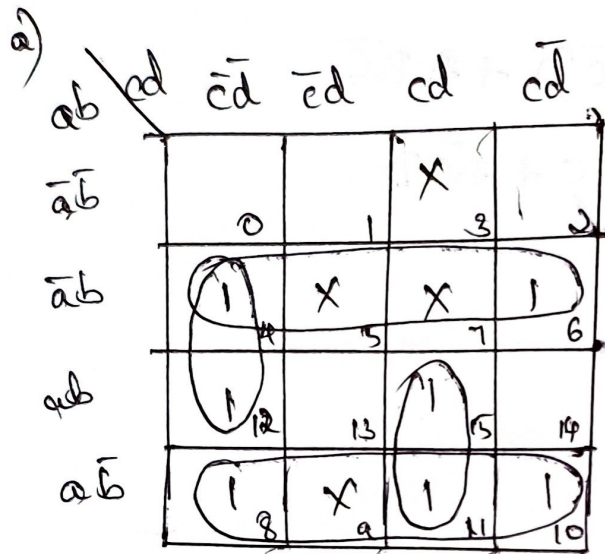
$$\underline{x_2 + x_3x_4 + \bar{x}_1\bar{x}_3}$$

4) Minimise the following Boolean expression using K-map.

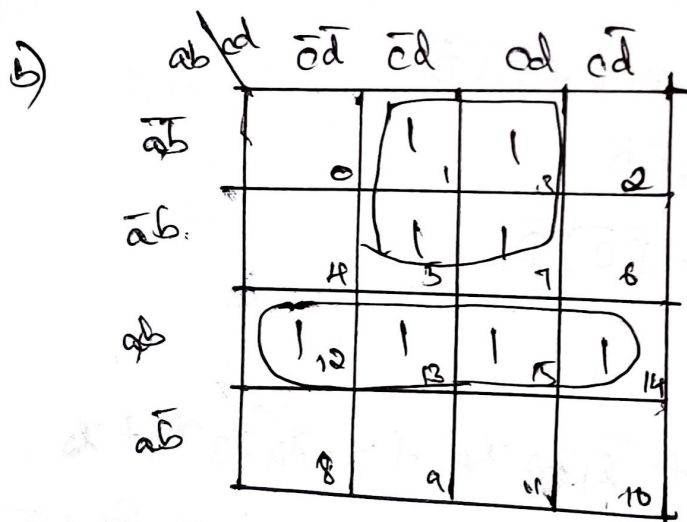
a)  $f(a, b, c, d) = \sum m(4, 6, 8, 10, 11, 12, 15) + \sum d(3, 5, 7, 9)$

b)  $f(a, b, c, d) = \prod m(0, 2, 4, 6, 7, 9) \cdot \prod d(10, 11)$

$\hookrightarrow \sum m(1, 3, 5, 7, 12, 13, 14, 15)$



$\bar{a}b + ab + b\bar{c}d + acd$



$ab + \bar{a}d$